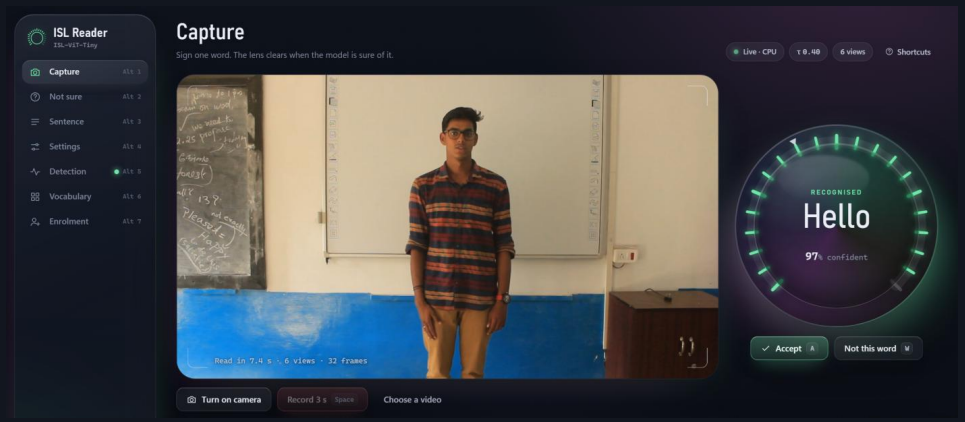
ISL Reader

A 2 MB Vision Transformer, a super-resolution GAN front end and an integrated interface for Indian Sign Language



Yalamanchi Kushal · Nethi Kushala Kumar · Dhruseek Rishi Menon
Guide: Dr. T. Bagyammal · 23CSE473 NNDL · Group 03 · Review 3

What we set out to build

A model of about 2 MB

It has to run on smart glasses, so the classifier must fit in roughly two megabytes.

≥ 75 % on unseen people

Measured only on recording sessions the model never saw, on the 50 deployed words.

A working app

Camera or video in, word out, with a confidence check and a way to correct mistakes.

Why it is hard

- Signs are two-handed: handshape, motion and location together decide the word.
- Little labelled data: about 15 clips per word in INCLUDE.
- A wearable camera gives smaller, compressed video than the 1080p studio data.
- Random splits leak recording sessions, so published accuracies overstate real use.

Dataset details

4,257
INCLUDE clips used

262
words, 15 categories

1080p
25 fps, one studio

13
recording sessions

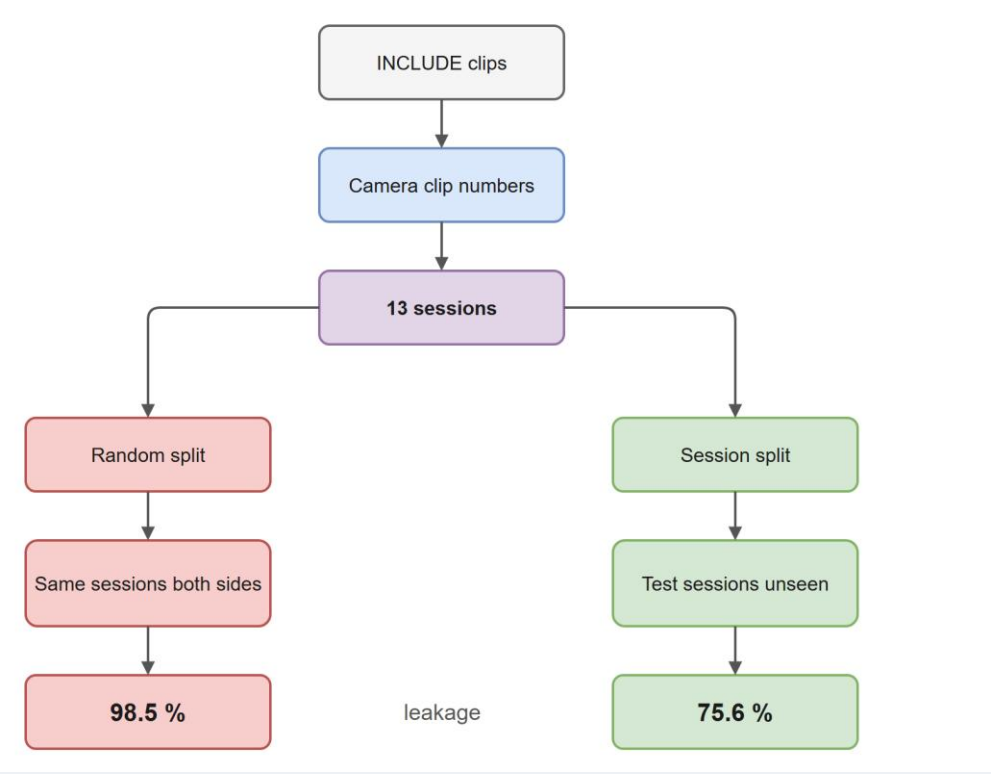
Split	Clips	Use
Training sessions	2,845	Classifier and GAN training
Held-out test sessions	1,010	472 of them = the 50 deployed words
Excluded	402	Words with < 6 training clips
Validation sessions	888	GAN development only
iSign (unlabelled)	18,000	Self-supervised pretraining
CISLR (300 x 300 px)	609	Cross-corpus test

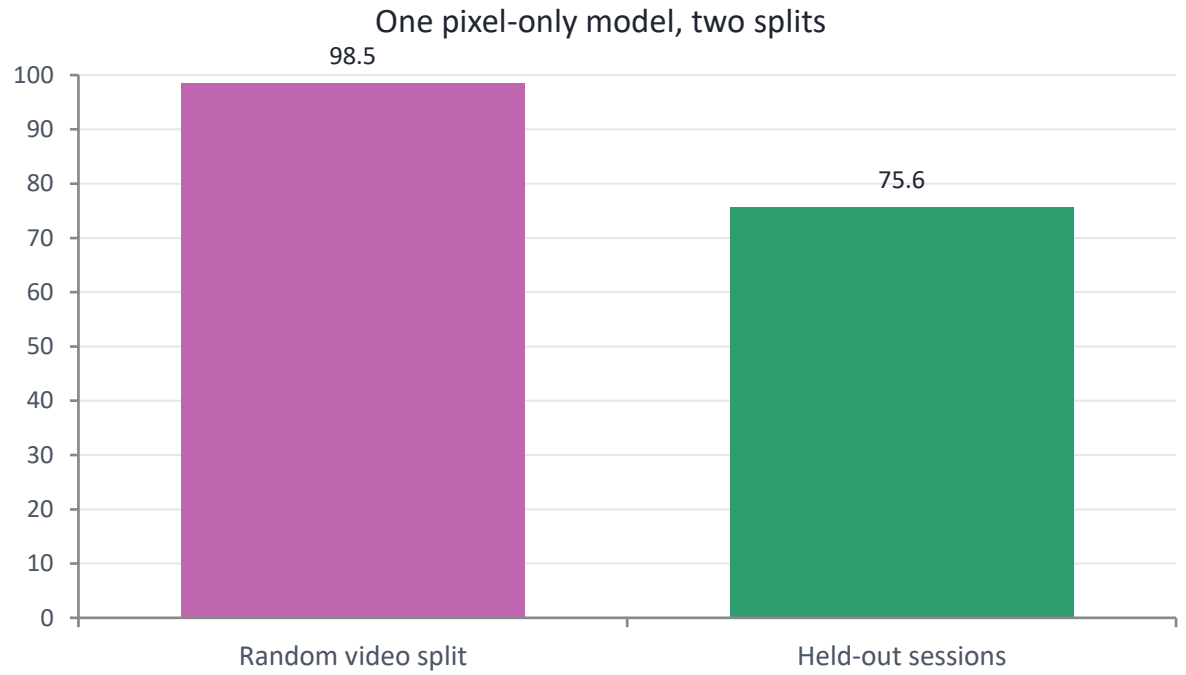


Frames from held-out INCLUDE clips (Sridhar et al., ACM MM 2020, CC-BY-4.0)

We train on all 262 words and answer from the 50 deployed words: per-word data is fixed, so a smaller training vocabulary only removes data.

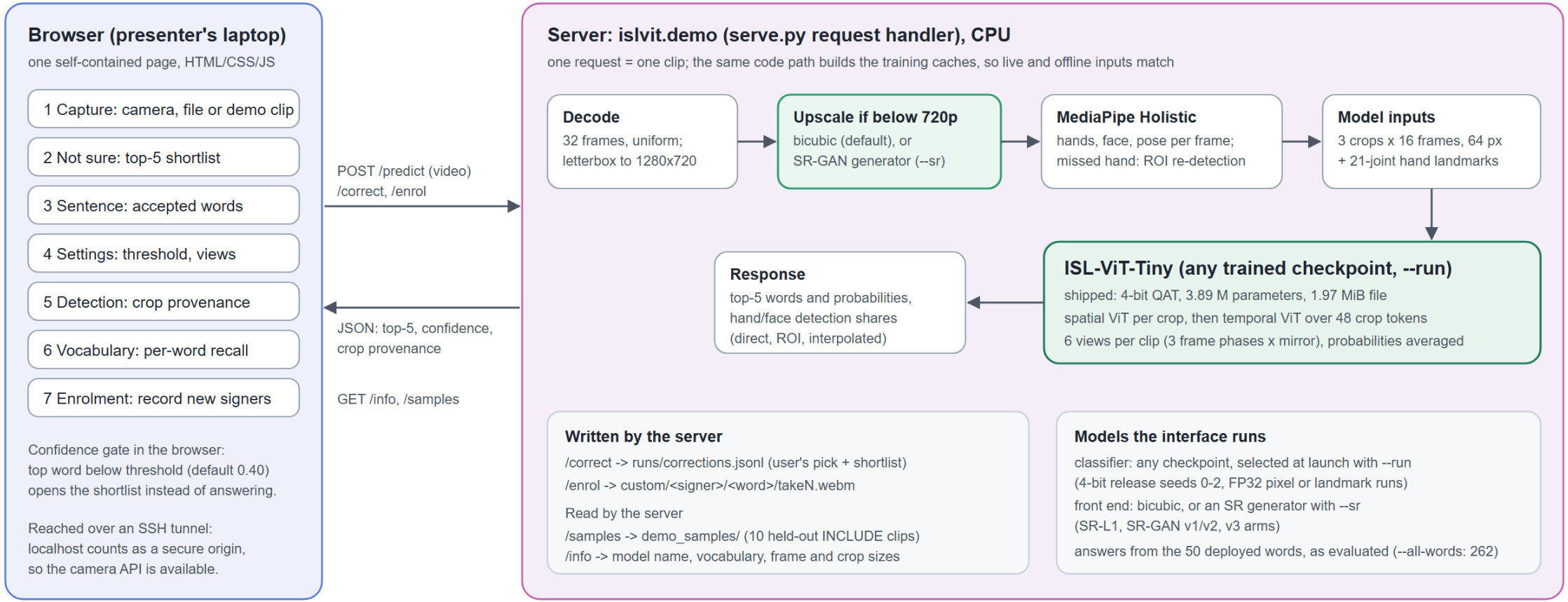
Evaluation: hold out whole recording sessions



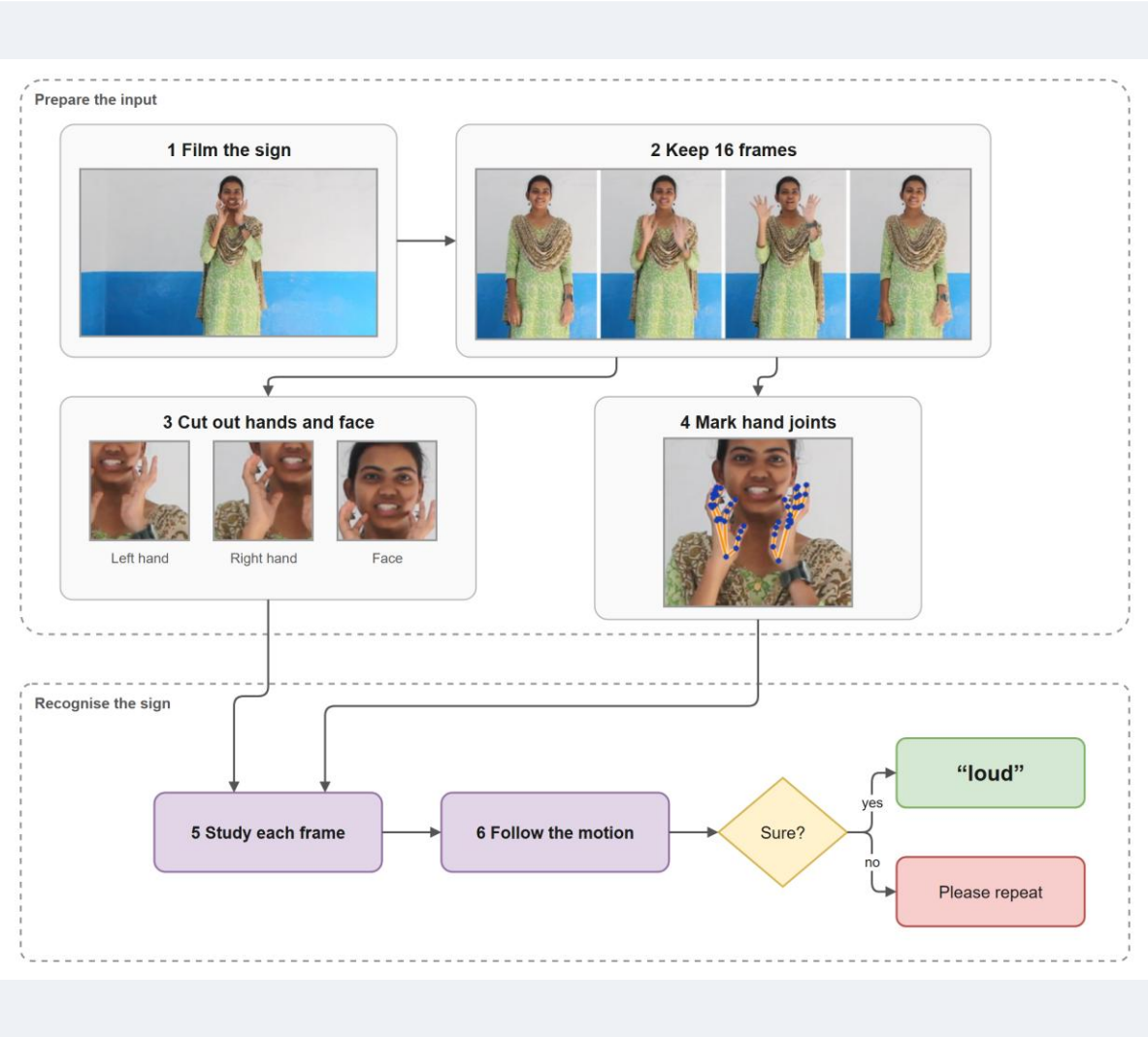


- Camera clip numbers run in sequence; big jumps mark a new session.
- Whole sessions are held out; a check refuses any overlap.
- The 22.9-point gap is what the model gains from seeing the same person and room.

System architecture of the integrated app

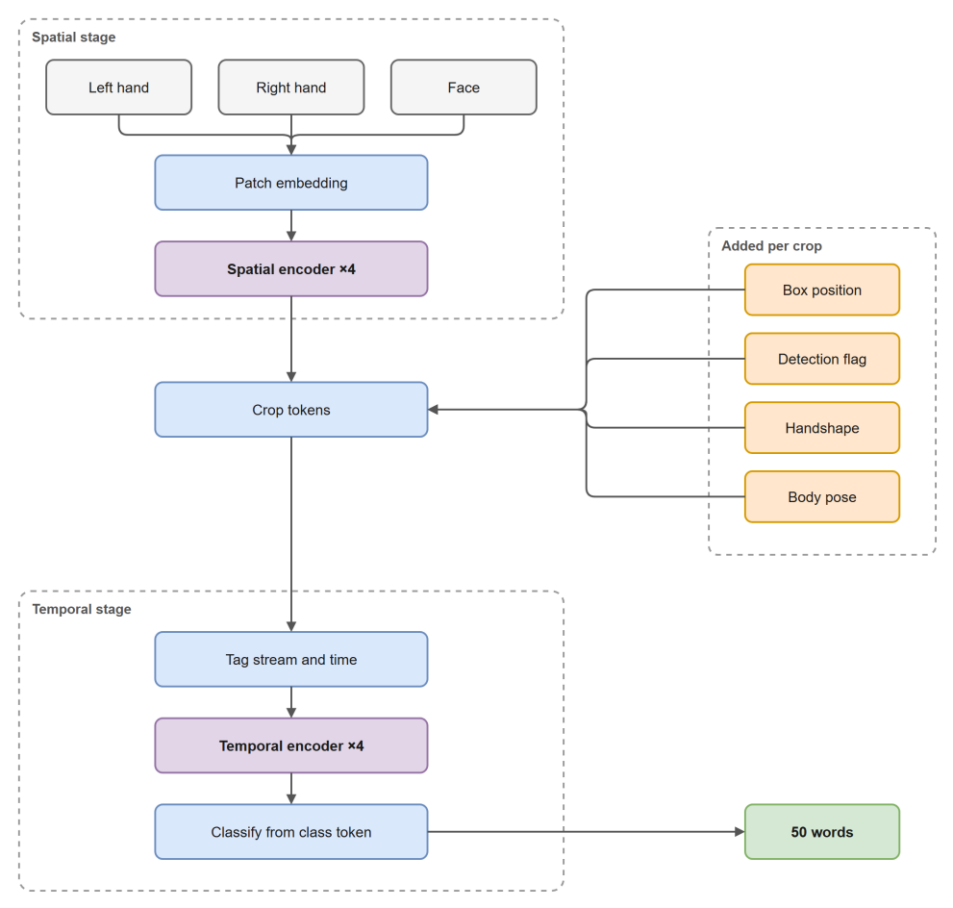


From video to model inputs



- 32 frames sampled uniformly, letterboxed to 1280 x 720.
- MediaPipe Holistic finds hands, face and pose; a missed hand gets a second search in an upscaled forearm region, else the nearest box is reused and flagged.
- Three crops per frame (left hand, right hand, face), 64 px.
- 21 joints per hand: shape relative to the wrist, location relative to the shoulders.
- Detectors are rebuilt for every video; otherwise repeated runs disagreed on 3–6 % of frames.

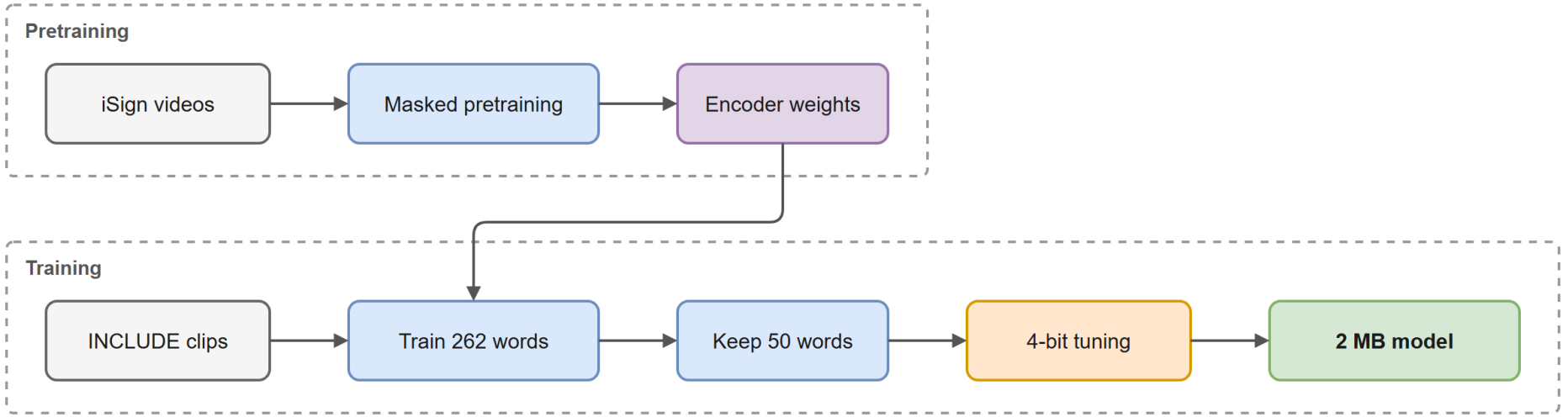
Classifier: ISL-ViT-Tiny



Component	Setting
Input	16 frames x 3 crops x 64 x 64, box geometry, detection flags, landmarks
Spatial stage	4 blocks, width 192, 3 heads, shared over 48 crops
Temporal stage	4 blocks over 48 crop tokens + class token
Output	262 words, restricted to the 50 deployed
Parameters	3,894,214 (91 k for landmarks)

Why factorised: joint attention over every patch would be 816 tokens; the model attends over 17 tokens per crop, then 49 across the clip. Box geometry puts back the location that cropping removes.

Training algorithm and compression



1. Self-supervised pretraining

Masked image modelling (SimMIM) on 18,000 unlabelled iSign clips: 75 % of patches hidden and reconstructed.

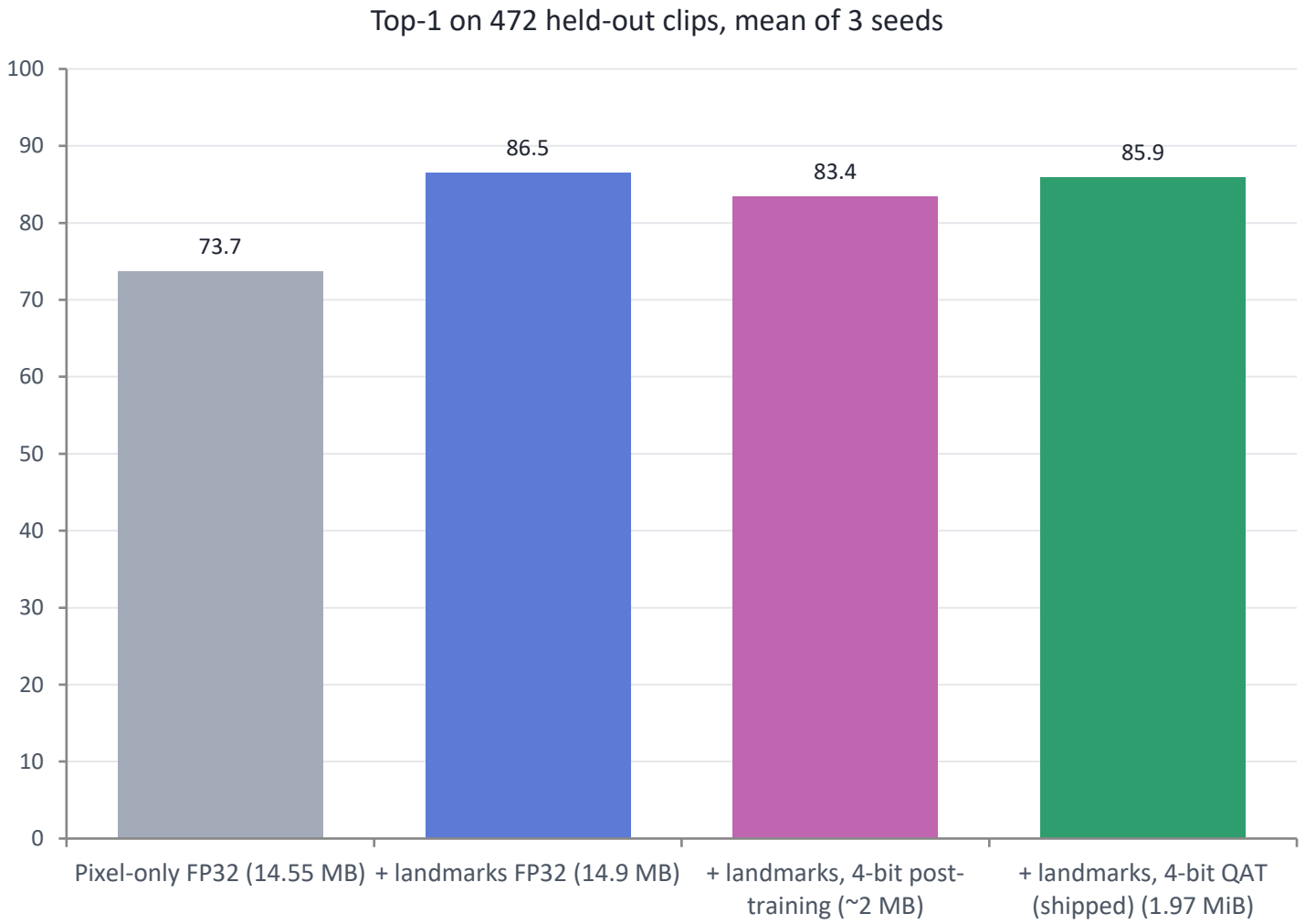
2. Supervised training

1,000 epochs on 2,845 clips, AdamW 5e-4, mixup, stream dropout, resolution jitter, EMA. The final epoch is kept: no selection on test.

3. 4-bit quantisation-aware

Weights at 4 bits in groups of 128 with 16-bit scales; straight-through estimator; 300 epochs. Post-training rounding lost 3.1 points, QAT only 0.6.

Comparing the classifier variants



85.9 %

top-1, shipped 4-bit model (seeds 84.5 / 87.5 / 85.8)

96.4 %

top-5: the right word is usually in the shortlist

1.97 MiB

3.89 M parameters, 4-bit

Answering from all 262 words instead of the 50 gives 72.0 %, so the app restricts answers exactly as the evaluation does.

Where a GAN can help: low-resolution video



Why here

Every clip reaches the hand detector through one resize to 720p, which was bicubic. A wearable camera would send small, compressed video through exactly this step. The GAN replaces only that resize; 1080p video never passes through it, so clean results cannot change.

How we tested it

472 test clips degraded to 360p, 240p, 180p with JPEG. Front ends: bicubic, SR-L1 (same generator, no adversarial loss), SR-GAN v1, v2 and two v3 arms. Frozen 4-bit and 32-bit models, plus models fine-tuned on 360p. Paired McNemar tests; a rule fixed before the runs.

3

levels: 360p, 240p, 180p, each with JPEG

6

front ends: bicubic, SR-L1, SR-GAN v1, v2, two v3 arms

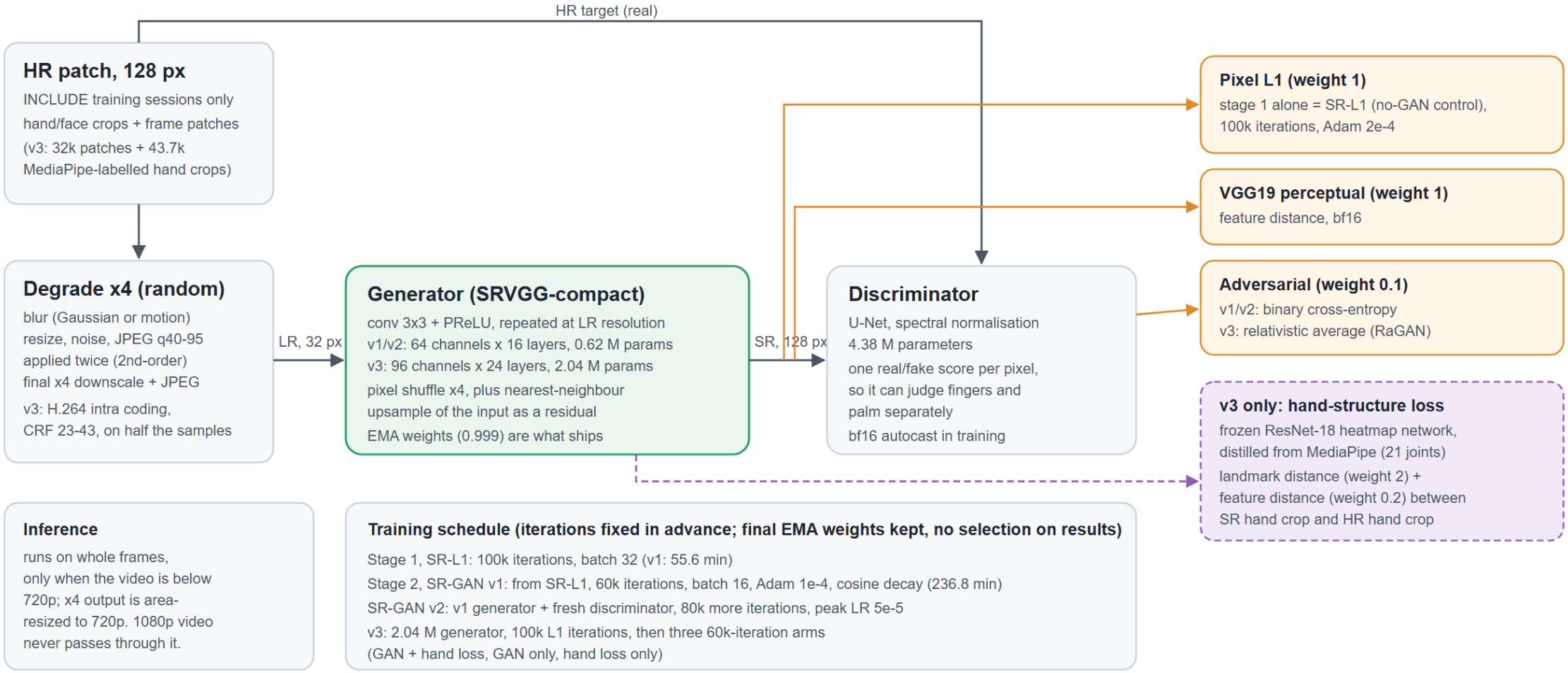
472

test clips, identical in every run

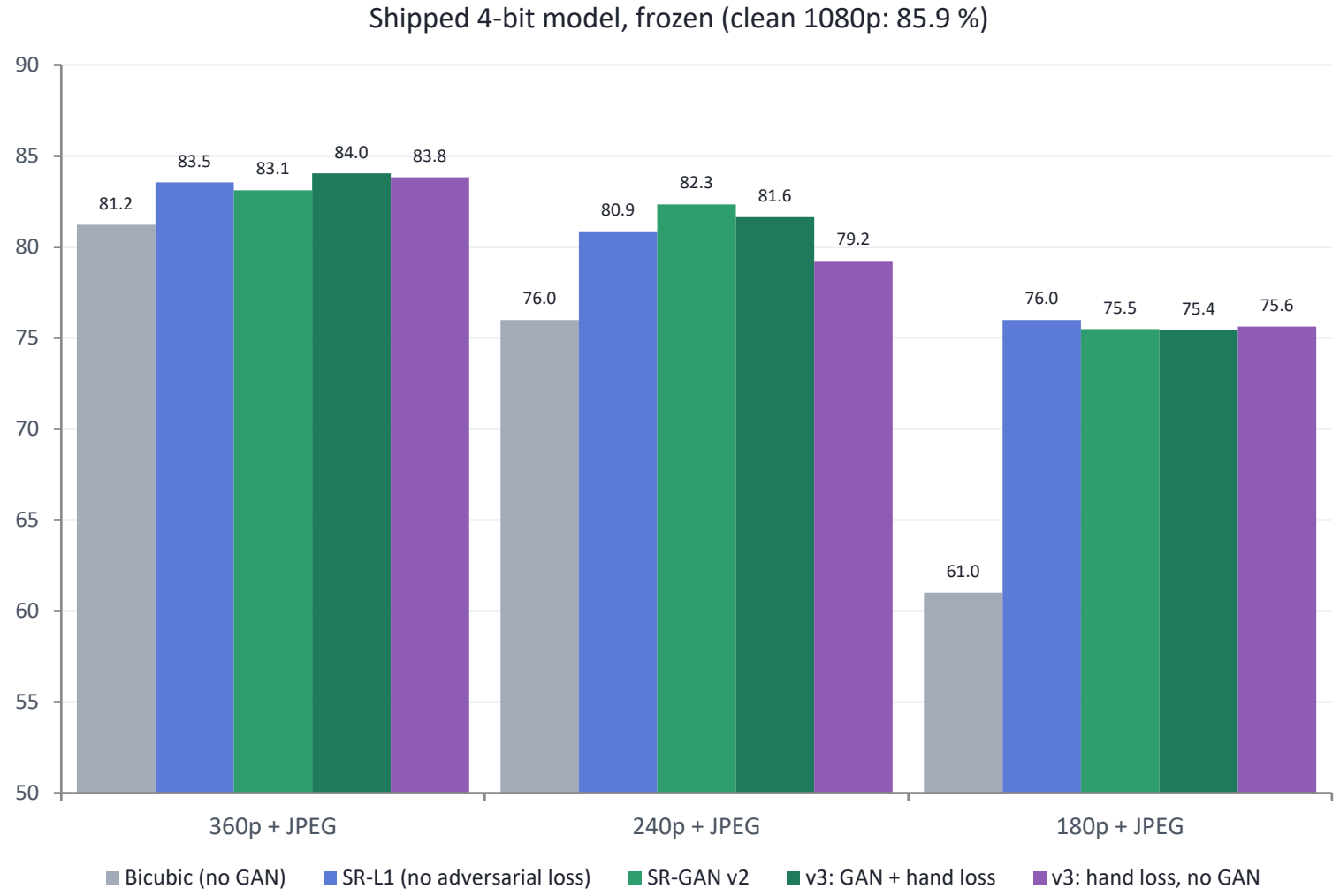
$p < 0.05$

and all 3 seeds agree, fixed in advance

The super-resolution GAN



GAN results on degraded test video



+14.5

points at 180p, SR-GAN v2 vs bicubic
($p = 7 \times 10^{-32}$)

+1.5

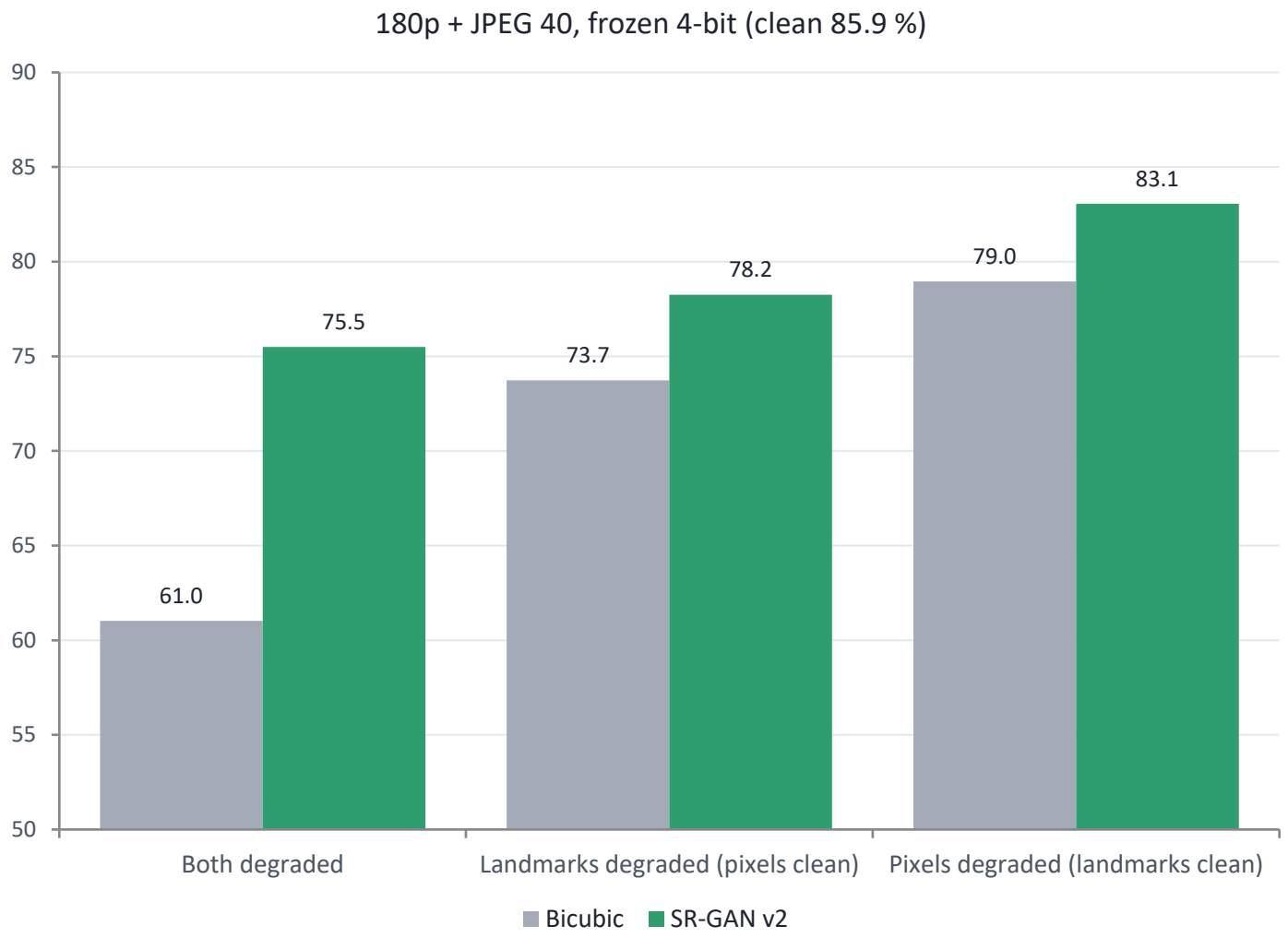
the adversarial loss's own gain over SR-L1, at 240p only

0

reliable gains for v3 over v2 or SR-L1, at any level

Fine-tuned on 360p: +5.2 at 240p, +12.5 at 180p over bicubic.

Where the remaining loss sits



Landmarks, not pixels

With SR-GAN, 10.4 points are still missing at 180p: about 7.7 from the landmarks and 2.8 from the pixels.

Detector tuning fails

Lower thresholds and wider searches found up to 99 % of hand frames, but accuracy moved at most 0.8 points: misplaced hands rose to 14 %.

No invented hands

Hands found only in degraded video: 1.9 % with bicubic, 2.6 % with SR-GAN, at 180p.

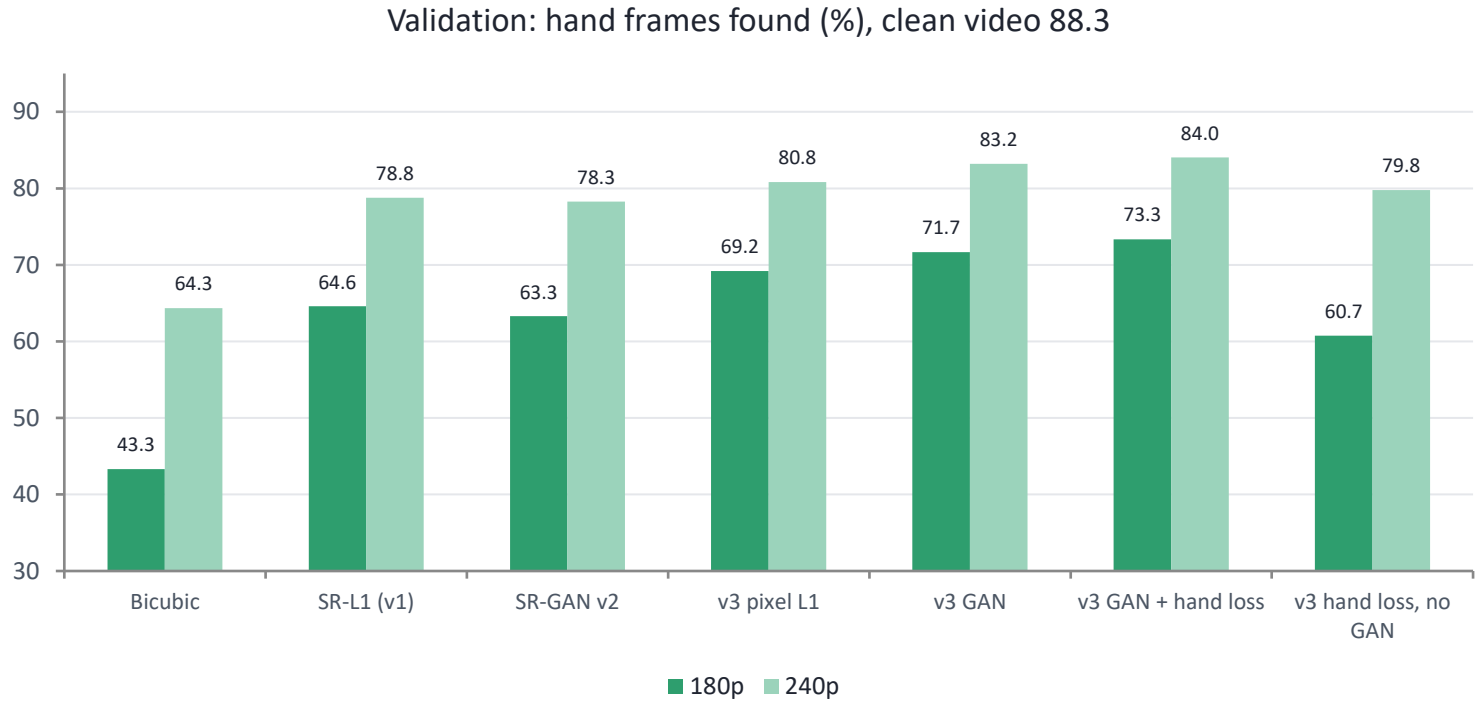
A task-aware GAN (v3): what it changed

Hand-landmark proxy

ResNet-18 with heatmap head, trained on 43,682 verified hand crops to read MediaPipe's 21 joints.

Error on validation crops: 0.0215 of the crop, 91.4 % of points within 0.05 (MediaPipe vs itself: 0.041).

The generator is penalised when the proxy reads different joints from its output than from the original.

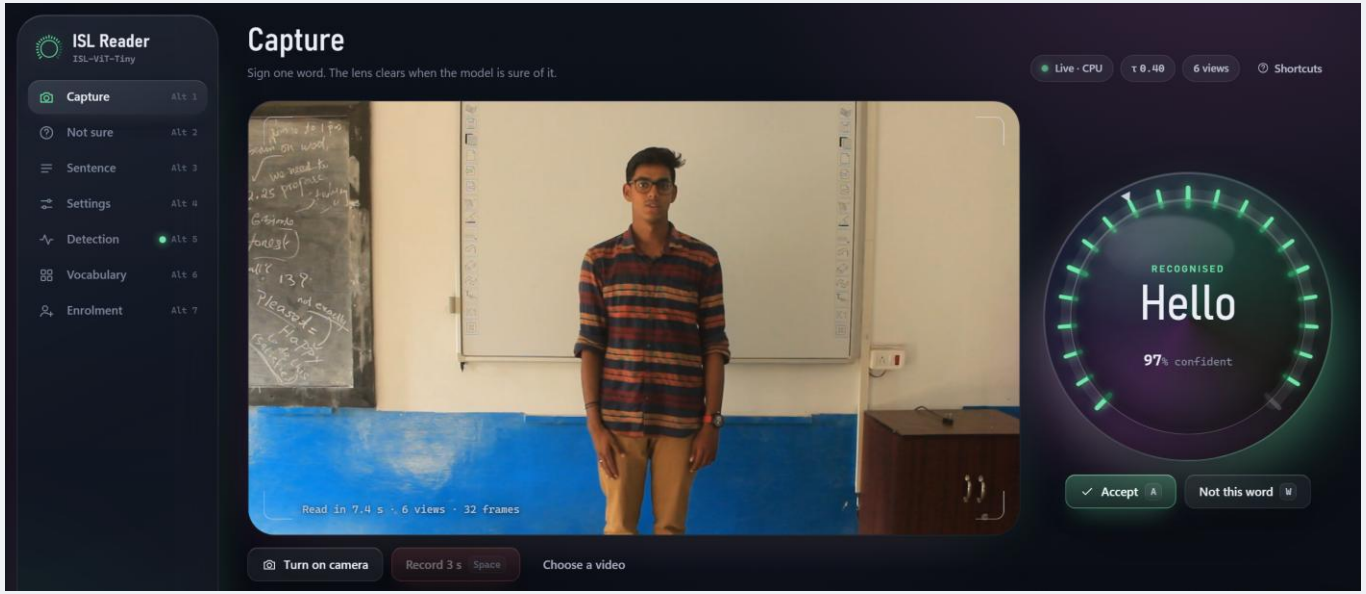


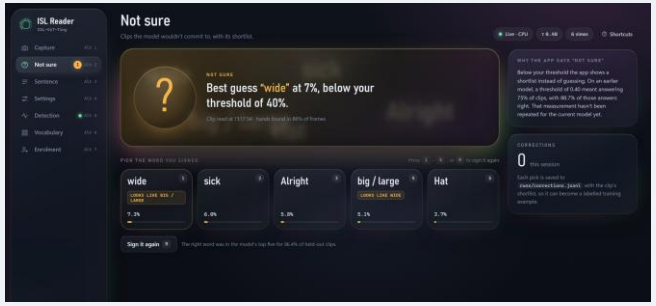
On validation it found more hands (73.3 % vs 63.3 % for v2 at 180p). On the test set it was not reliably better than v2 or SR-L1, and the version without the adversarial loss was 3.1 points worse at 240p. Better detection did not become better classification.

−3.1

points: v3 without the adversarial loss vs v2 at 240p ($p = 5 \times 10^{-6}$)

The interface, connected to the models







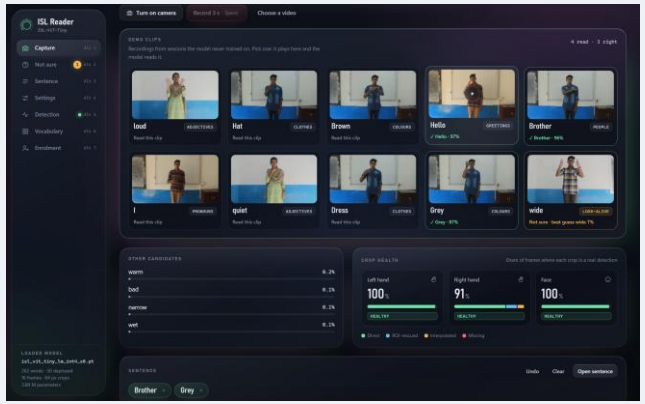
Seven screens

- Capture: camera, video file or one of 10 held-out demo clips
- Not sure: shortlist when confidence < threshold
- Sentence: accepted words, speak, copy, export
- Settings: threshold and views
- Detection: where each crop came from
- Vocabulary: per-word recall on unseen sessions
- Enrolment: record new signers for training

Runs any trained checkpoint and either upscaling front end; reached over an SSH tunnel, which also enables the camera.

Testing the interface with the models

Test	Method	Result
Screens render	Chromium, 7 screens at 1366 and 1920 px	No console errors
Live answers	10 demo clips on the live server; 4 also via the Capture screen	9/9 reliable clips correct (66–97 %); look-alike → Not sure
Same answers as the evaluation	App vs offline result, same clips and model	8 of 8 clips agree
Vocabulary restriction	50-word vs 262-word answers	Matches the 85.9 % setting (262 words: 72.0 %)
GAN front end in the app	180p copy of a demo clip, SR-GAN on	Read correctly
Latency	Desktop CPU, 6 views, end to end	5–9 s per clip





What we found

- 85.9 % top-1 on unseen sessions from a 1.97 MiB model; random splits overstate accuracy by over 20 points.
- Learned super-resolution recovers up to 14.5 points on 180p video; the adversarial loss adds little beyond it.
- After upscaling, the remaining error is in landmark placement, not pixels.
- The app reproduces the evaluation's answers clip for clip.

Next steps

- Fine-tune the classifier on the task-aware front end
- Re-measure the confidence curve for the shipped model
- Fix hand detection for tightly framed video (CISLR)
- Record new signers through Enrolment; test on wearable hardware